

Test 2 — Question Paper

38 items. Suggested time: 95 minutes (about 2 minutes 30 seconds per item).

A four-function calculator may be used **only** on items marked *calculator*.

The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.

Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. What is the value of 1.2^2 ? Enter your answer.

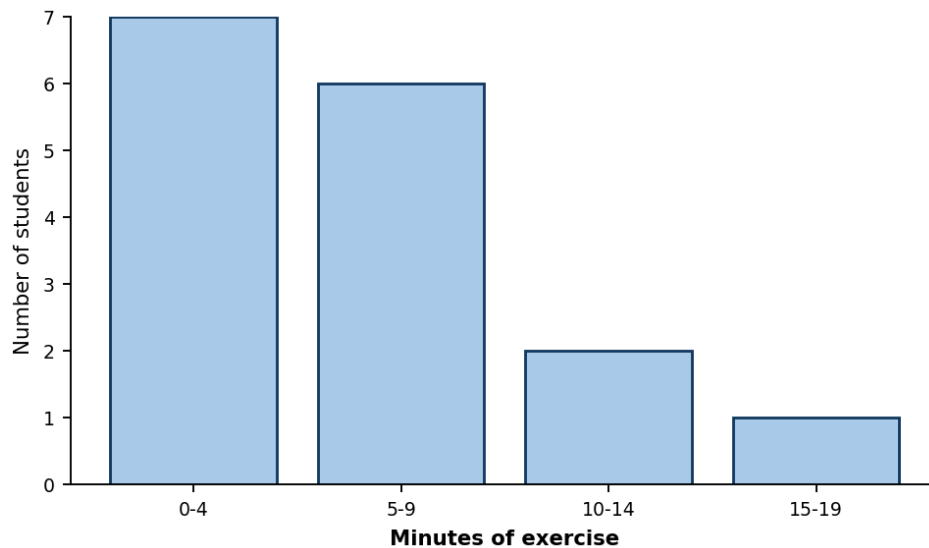
Answer: _____

Show your work

calculator · equation editor

2. The histogram shows the data collected from a group of students.

Which statement best describes the shape of the data?



- A. The data are symmetric, with about the same number of values on each side.
- B. The data have no values in the lowest interval.
- C. The data are skewed to the left, with most values in the higher intervals.
- D. The data are skewed to the right, with most values in the lower intervals.

Show your work

multiple choice

3. On a January morning the temperature in Homestead was 5 degrees below zero Celsius at a research freezer.

That temperature is °C, and its opposite is °C.

Blank (1) choices: -5 | 5 | -10 | 0

Blank (2) choices: -5 | 5 | -10 | 0

Show your work

calculator · editing task choice

4. A recipe uses 16 cups of rice and 10 cups of water.

Write the ratio of rice to water in simplest form. Use the form a:b.

Answer:

Show your work

equation editor

5. The table shows equivalent ratios of juice to water.

Enter the number that belongs in the box marked ?

Cups of juice	Cups of water
5	6
25	?

Answer:

Show your work

equation editor

6. Select **all** the numbers whose absolute value is 8.

Circle every correct choice:

- ☐ 8
- ☐ -8
- ☐ 16
- ☐ -16
- ☐ 0
- ☐ 9

Show your work

calculator · selectable hot text

7. The table shows the points scored in five games.

What is the mean number of points? Enter your answer.

Points	#1	#2	#3	#4	#5
Points	17	22	23	25	33

Answer: _____

Show your work

equation editor

8. Camila needs $1\frac{1}{2}$ cups of flour for one batch of pastelitos.

How many cups are needed for 4 batches? Enter your answer as a fraction or a mixed number.

Answer: _____

Show your work

calculator · equation editor

9. A teacher has 24 pencils and 36 erasers to put into identical bags with nothing left over.

What is the greatest number of bags she can make?

Answer: _____

Show your work

calculator · equation editor

10. A fence is 6 yards long.

How many feet long is the fence? Enter your answer. (1 yard = 3 feet)

Answer: _____

Show your work

equation editor

11. Select **all** the questions that are statistical questions.

- ☐ How many text messages did each student send yesterday?
- ☐ How many minutes did each sixth grader spend on the bus this morning?
- ☐ How old is my teacher?
- ☐ What time does the first bell ring at my school?
- ☐ How many books did I read last month?

Select all that apply.

Show your work

multiselect

12. A store sells guavas for \$2.40 each.

What is the total cost of 20 of them? Enter your answer in dollars.

Answer: _____

Show your work

calculator · equation editor

13. A rideshare charges a base fare of \$9.00 plus \$4.00 per mile.

What is the cost, in dollars, of a ride of $x = 2$ miles? Enter your answer.

Answer: _____

Show your work

equation editor

14. Use the drop-down menu to make the statement true.

$$|-17| \text{ } [[1]] \text{ } |4|$$

Blank (1) choices: < | > | =

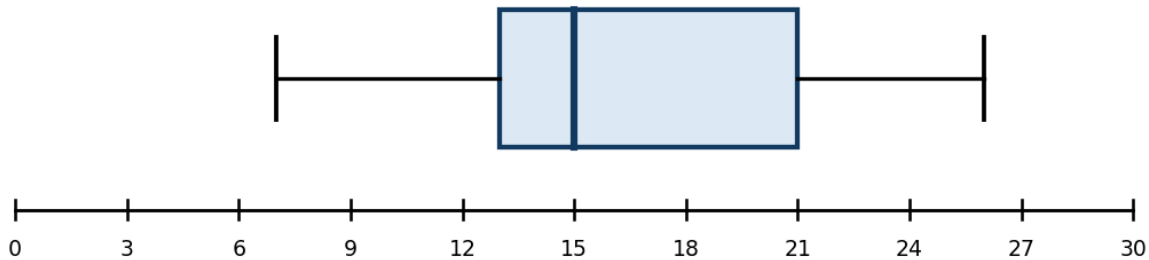
Show your work

calculator · editing task choice

15. The box plot shows a set of data.

What is the maximum? Enter your answer.

Daily high temperature (°F)



Answer: _____

Show your work

equation editor

16. Use the drop-down menus to complete the equivalent expression.

$$8(6x + 6) = [[1]]x + [[2]]$$

Blank (1) choices: 6 | 14 | 48 | 54

Blank (2) choices: 6 | 14 | 48 | 54

Show your work

editing task choice

17. At 6 a.m. the temperature was -12°F in a walk-in freezer. It rose 16°F and then fell 10°F .

What is the temperature now, in $^{\circ}\text{F}$? Enter your answer.

Answer: _____

Show your work

calculator · equation editor

18. A pair of headphones costs \$60.00 and is on sale for 50% off.

What is the sale price, in dollars? Enter your answer.

Answer: _____

Show your work

equation editor

19. Which number is located between 1 and 2 on a number line?

- A. $1\frac{1}{4}$
- B. 3
- C. $2\frac{1}{4}$
- D. $\frac{1}{2}$

Show your work

calculator · multiple choice

20. A pack of 12 bottles of water costs \$58.68.

What is the cost per bottle, in dollars? Enter your answer.

Answer: _____

Show your work

equation editor

21. A pond loses 8 centimeters of water each week during a dry spell.

Which integer represents the total change in water level after 5 weeks? Enter your answer.

Answer: _____

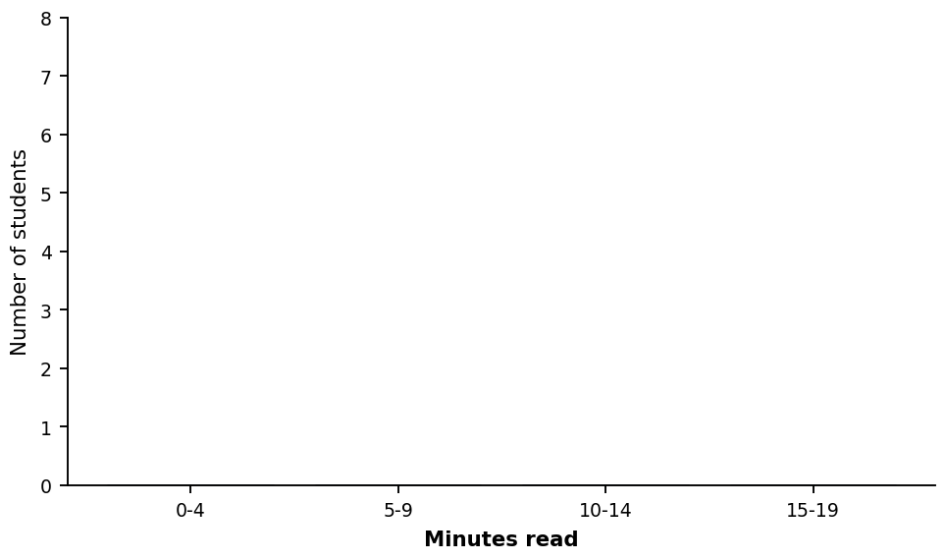
Show your work

calculator · equation editor

22. A data set is shown.

1, 2, 2, 4, 4, 4, 7, 10, 10, 11, 11, 17

Build the histogram by setting the height of each bar.



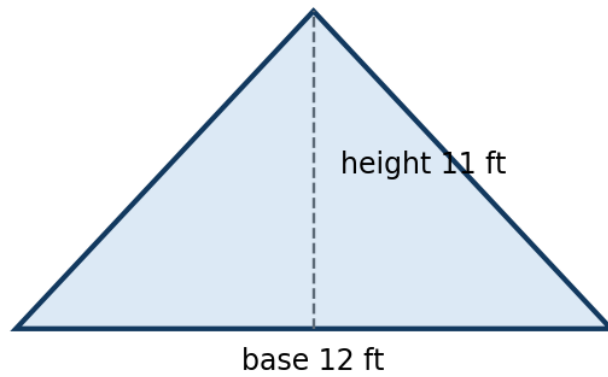
Mark your answer on the diagram above.

Show your work

graphing (grid)

23. A triangular sail has the base and height shown.

What is the area of the sail, in square feet? Enter your answer.



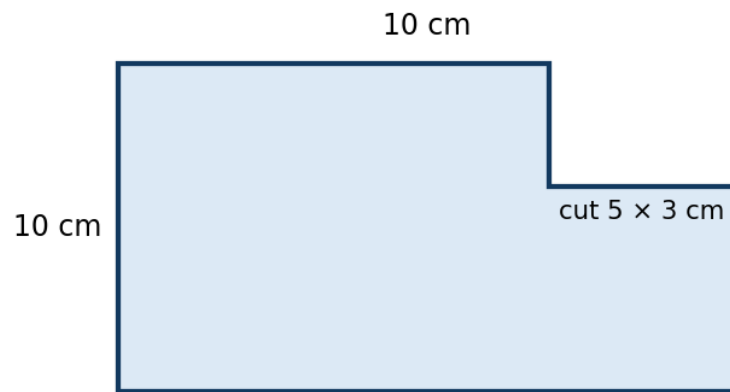
Answer: _____

Show your work

equation editor

24. The figure shows the floor of a room. All corners are right angles.

What is the area of the floor, in square centimeters? Enter your answer.

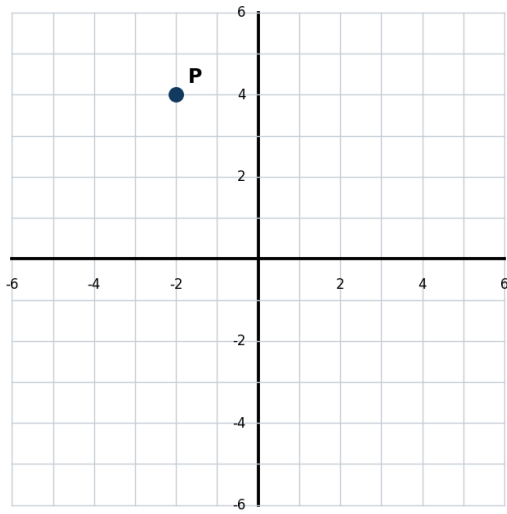


Answer: _____

Show your work

equation editor

25. What are the coordinates of point P?



- A. (2, 4)
- B. (-2, -4)
- C. (-2, 4)
- D. (4, -2)

Show your work

multiple choice

26. What value of x makes the equation $x \div 4 = 9.74$ true? Enter your answer.

Answer: _____

Show your work

equation editor

27. Solve the equation $x \div 6 = 5$ for x . Enter your answer.

Answer: _____

Show your work

equation editor

28. A roll holds 2 yards of ribbon. Marcos cuts 14 pieces that are each $\frac{3}{4}$ yard long.

How many yards of ribbon are left? Enter your answer as a fraction or a mixed number.

Answer: _____

Show your work

calculator · equation editor

29. Select **all** the values that make $6x + 9 = 75$ true.

Circle every correct choice:

- ☐ 9
- ☐ 11
- ☐ 14
- ☐ 18
- ☐ 75

Show your work

selectable hot text

30. What is the greatest common factor you would use to rewrite $36 + 84$ as a product? Enter your answer.

Answer: _____

Show your work

calculator · equation editor

31. Which expression represents the sum of a number x and 10, divided by 2?

- A. $x/2 + 210$
- B. $2(x + 10)$
- C. $x + 10/2$
- D. $(x + 10)/2$

Show your work

multiple choice

32. What is the opposite of $-8 \frac{1}{2}$?

- A. $8 \frac{1}{2}$
- B. $4 \frac{1}{4}$
- C. $-8 \frac{1}{2}$
- D. -17

Show your work

calculator · multiple choice

33. Solve the equation $x - 25 = 31$ for x . Enter your answer.

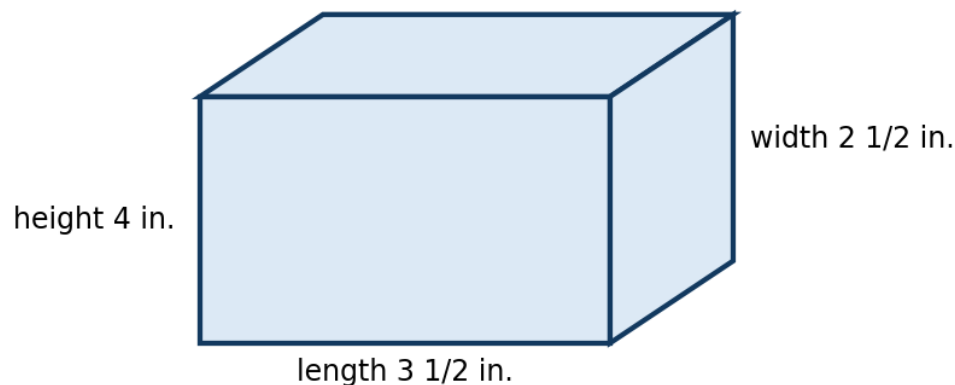
Answer: _____

Show your work

equation editor

34. A storage bin is a right rectangular prism with the dimensions shown.

What is its volume, in cubic inches? Enter your answer as a fraction, a mixed number or a decimal.



Answer: _____

Show your work

equation editor

35. Enter the prime factorization of 84. Use ^ for exponents and $\times$ between factors, for example $2^2 \times 3$.

Answer: _____

Show your work

calculator · equation editor

36. Write 0.6 as a percent. Enter your answer without the % symbol.

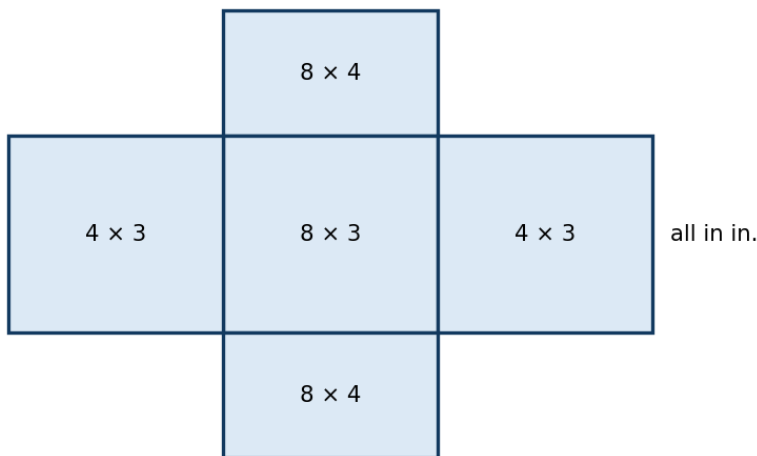
Answer: _____

Show your work

calculator · equation editor

37. A gift box is a right rectangular prism measuring 8 by 4 by 3 inches. Its net is shown.

How many square inches of paper are needed to cover the box exactly? Enter your answer.



Answer: _____

Show your work

equation editor

38. A data set is shown.

6, 8, 12, 22, 25, 27, 28

The value 6 is removed. What is the median of the new data set? Enter your answer.

Answer: _____

Show your work

equation editor

Grade 6 Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches	1 cup = 8 fluid ounces
1 yard = 3 feet	1 pint = 2 cups
1 mile = 5,280 feet	1 quart = 2 pints
1 mile = 1,760 yards	1 gallon = 4 quarts
1 pound = 16 ounces	1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters	1 liter = 1,000 milliliters
1 meter = 1,000 millimeters	1 gram = 1,000 milligrams
1 kilometer = 1,000 meters	1 kilogram = 1,000 grams

Time Conversions

1 minute = 60 seconds	1 week = 7 days
1 hour = 60 minutes	1 year = 365 days
1 day = 24 hours	1 year = 52 weeks

Formula

Rectangular prism	$V = lwh$ or $V = Bh$
l = length, w = width, h = height	B = area of base, V = volume